Additional file 3: Detailed inclusion and exclusion criteria

The detailed inclusion and exclusion criteria listed here is for the overall CDS systematic review, a mixed-method review incorporating qualitative, effectiveness, and economic evaluation components. Only the qualitative outcomes in the inclusion criteria apply to this study.

Inclusion criteria
Population: CDS must be for adults (>18 years) receiving chronic disease care: 1. Chronic kidney disease: including monitoring of acute kidney injury in community settings. 2. Cardiovascular disease: including ischaemic heart disease, cerebrovascular disease, peripheral vascular disease, and abdominal aortic aneurysm. Includes diagnosis of chest pain and transient ischaemic attack in community settings. 3. Diabetes: including type 1 diabetes mellitus, type 2 diabetes mellitus, gestational diabetes mellitus. 4. Hypertension 5. Hypercholesterolaemia 6. Other cardiovascular risk factors: metabolic syndrome, obesity.
Intervention: EHR-enabled, computerised CDS used in any non-acute setting. CDS includes and is not limited to: 1. CDS targeting chronic disease care: for screening, diagnosis, pathway support, pharmacological management, non-pharmacological management. 2. CDS functions: including tools for clinical documentation, data presentation, order or prescription creation, protocol or pathway support, reference guidance, and alert or reminder. 3. CDS used in any setting for chronic disease care: including primary care, specialist outpatient, and community health services. May include registries and other databases derived from EHR if the system includes an EHR-based CDS.
Comparison: Comparison can be a control group receiving placebo or standard care, or a historic control for effectiveness outcome studies only.
Outcomes: Outcomes include: 1. Effectiveness outcomes: including clinical effectiveness (e.g. morbidity, mortality), health process outcomes (e.g. adherence to guidelines), workload and efficiency outcomes (e.g. time saved). 2. Economic outcomes: including cost, cost-effectiveness, cost-utility, cost-benefit outcomes. 3. Qualitative outcomes: experience of healthcare providers (individual clinicians or services) in implementing, using, evaluation and sustaining CDS interventions.
Other eligibility criteria: Time restricted to publications from January 2011 to January 2021. English language studies only.

Exclusion criteria
Population: CDS for paediatric populations (<18 years). CDS were excluded if they targeted patients: 1. With acute conditions: e.g. deep vein thrombosis, pulmonary embolism, acute kidney injury in hospital setting, communicable diseases. 2. Within acute settings: e.g. intensive care unit, surgical, inpatient settings.
Intervention: CDS interventions were excluded if they were:

1. CDS without an EHR component: e.g. manual paper-based CDS without an EHR component, CDS in a stand-alone web-based application, CDS as a component of patient education module without links to EHR records, telehealth with CDS component without link to EHR. 2. EHR without a CDS intervention: e.g. implementation of new EHR. 3. Electronic monitoring interventions without an EHR component: e.g. biosensor or biological monitoring, robotics, ECG monitoring. 4. CDS with limited scope for chronic disease care: e.g. sole focus on image interpretation (CT diagnoses, retinal screening), sole focus on single medication dosing (insulin, warfarin dosing), simple alert or reminder (EHR-generated patient recall alerts).
Comparison: Studies without a control group for effectiveness outcomes only.
Outcomes: No exclusions based on outcome.
Other exclusion criteria: Excluded studies: 1. Publication: before 2011, non-English language. 2. Article type: not primary research articles (e.g. perspective pieces, systematic reviews, editorials), or were protocols of primary research studies only. 3. CDS development or system in test conditions only: e.g. modelling or technical methods–risk prediction, AI data mining, model performance, ontology development. 4. CDS testing in test conditions only: prototype not used in the clinical setting.

Abbreviations:

AI – Artificial intelligence

CDS – Clinical decision support

ECG – Electrocardiogram

EHR – Electronic health record